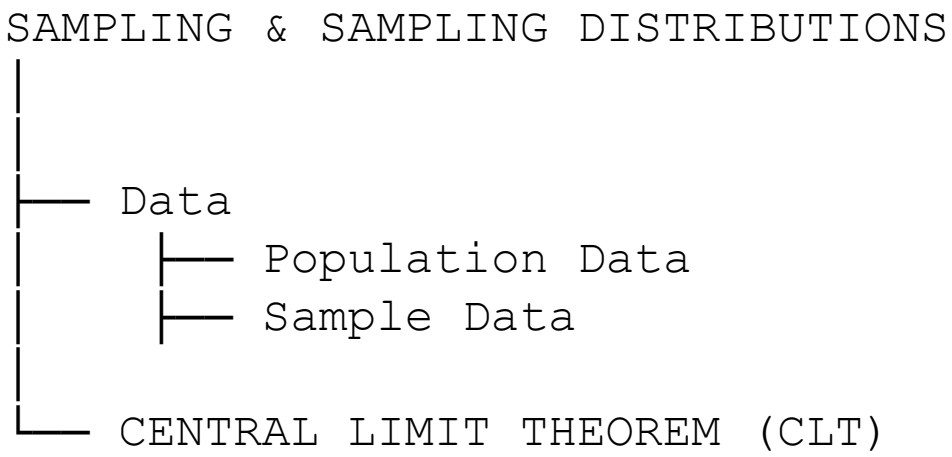
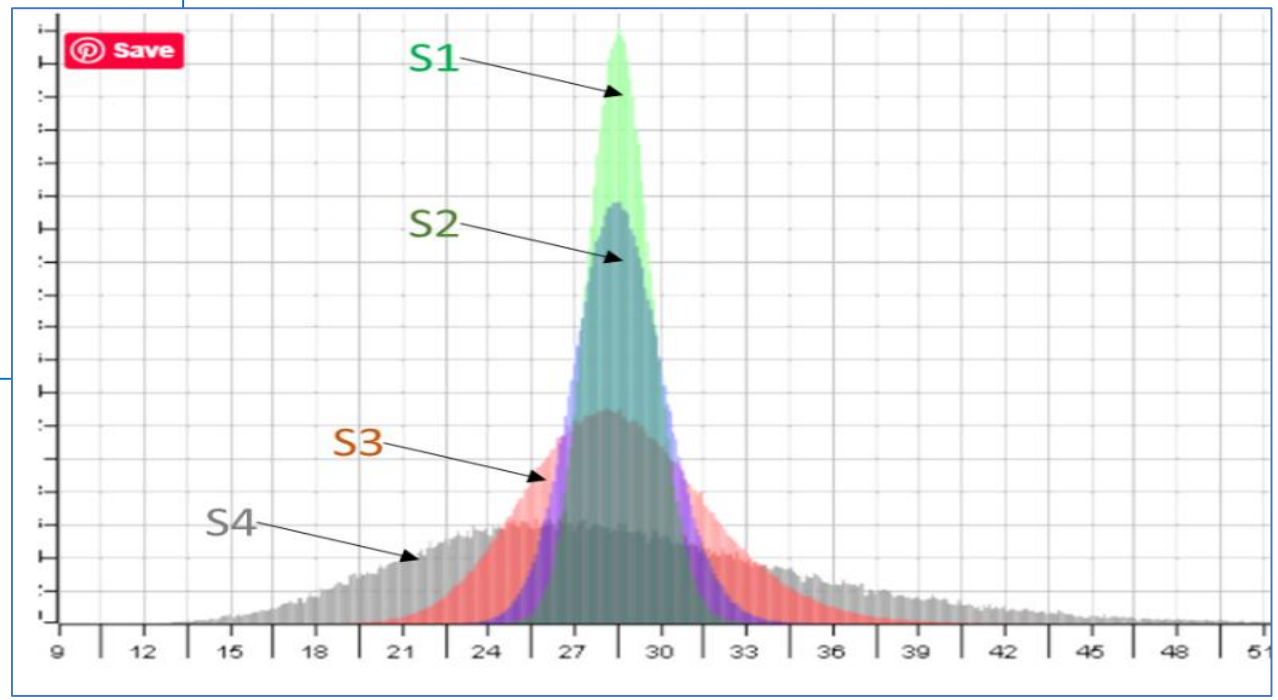

SAMPLING & SAMPLING DISTRIBUTIONS
Data
Population Data
Sample Data
CENTRAL LIMIT THEOREM (CLT)

Central Limit Theorem:

If you repeatedly take random samples of size n from any population and calculate the sample mean, the distribution of those sample means will approach a **normal distribution (bell curve)** as the sample size becomes large, regardless of the shape of the original population.

Number of Sample = m
Size of Sample = n

			Mean of Each Sample
S_1	{X1,X2,X3,X4,.....Xn}	------->	$\bar{X}_1$
S_2	{X1,X2,X3,X4,.....Xn}	------->	$\bar{X}_2$
S_3	{X1,X2,X3,X4,.....Xn}	------->	$\bar{X}_3$
S_4	{X1,X2,X3,X4,.....Xn}	------->	$\bar{X}_4$
			⋮
			$\bar{X}_m$



1.Take many random samples of size 30.
2.Compute the mean of each sample.
3.Plot all those means.
The resulting distribution of means will look approximately like a bell curve(Normal Distribution).

What Happens to the Mean and Spread?

According to CLT:

Mean of sample means = population mean (μ)

Standard deviation of sample means =

This quantity is called the **Standard Error (SE)**.

As sample size increases:

Standard error decreases

Estimates become more precise

$$SE = \frac{\sigma}{\sqrt{n}}$$

← Standard deviation

← Number of samples